

# LLM-Assisted Multi-Agent Stock Decision Support System

## Final Course Project Presentation

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Representation and Reasoning: Foundations of Large Models

Final Presentation

# Problem and Goal

- Build an applied prototype for daily-frequency simulated stock decision support.
- Support three tasks: single-stock analysis, candidate screening, and portfolio rebalancing.
- Avoid a black-box “ask the LLM for a stock pick” workflow.
- Produce reproducible outputs, explicit risk warnings, and benchmarkable behavior.

# System Workflow

| Data Layer                                    | Agent Layer                                       | Decision Layer                                                | Evaluation Layer                                  |
|-----------------------------------------------|---------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------|
| Prices, news,<br>fundamentals,<br>constraints | Market, sentiment,<br>fundamental, risk<br>agents | Decision synthesis<br>plus optional<br>OpenRouter<br>reviewer | Fixed cases,<br>baselines, backtest,<br>dashboard |

## Design principle

The LLM sees structured intermediate evidence and reviews the decision; deterministic fallback keeps the benchmark reproducible.

## Signal agents

- MarketInformationAgent
- NewsSentimentAgent
- FundamentalAgent

## Control agent

- RiskManagerAgent checks volatility, drawdown, concentration, cash, and trade limits.

## Decision agents

- DecisionAgent combines weighted evidence.
- LLMDecisionAdvisor optionally adjusts the result through OpenRouter JSON review.

## Output

Action, score, confidence, rationale, risk warnings, target weights, and trade deltas.

# Data and Benchmark

- Offline processed dataset: 12 large-cap tickers, 60 trading days.
- Inputs: OHLCV price history, sample news events, fundamentals, and portfolio constraints.
- Final benchmark: 15 fixed cases.

| Task                  | Cases | Purpose                                |
|-----------------------|-------|----------------------------------------|
| Single-stock analysis | 10    | action, rationale, risk awareness      |
| Candidate screening   | 2     | ranked stock pool selection            |
| Rebalancing           | 3     | portfolio constraints and trade deltas |

# Evaluation Results

| Metric                         | Result                       |
|--------------------------------|------------------------------|
| Unit tests                     | 6/6 passing                  |
| Final benchmark cases          | 15/15 passing                |
| OpenRouter smoke test          | passed with gpt-oss-20b free |
| Multi-agent risk warnings      | 16                           |
| Equal-weight cumulative return | 4.77%                        |
| Equal-weight max drawdown      | -1.37%                       |
| Equal-weight Sharpe ratio      | 2.44                         |

## Interpretation

The benchmark evaluates structure, constraints, and reproducibility. It is not a claim of live trading profitability.

# Demo Plan

## ① Run the final benchmark:

```
.\.venv\Scripts\python.exe scripts\run_demo.py --task all --llm off
```

## ② Open generated outputs:

- reports/final\_benchmark\_results.md
- reports/final\_dashboard.html

## ③ Optional LLM mode:

```
$env:OPENROUTER_API_KEY="..."
```

```
.\.venv\Scripts\python.exe scripts\run_demo.py --task final --llm required
```

- Runnable Python package under `src/stock_agent/`.
- CLI entry point: `scripts/run_demo.py`.
- Deterministic data generator: `scripts/generate_final_data.py`.
- Automated tests under `tests/`.
- Reports: Markdown summary and static HTML dashboard.
- Final written report and this presentation deck.



## Limitations and Next Steps

- Current data is deterministic and offline; live market/news feeds are future work.
- Scoring rules are transparent but simple; calibrated statistical models could improve them.
- The backtest is auxiliary and equal-weight, not an optimized trading strategy.
- Next steps: richer demo video, broader stock universe, and more realistic transaction-cost modeling.

# Takeaway

## Final outcome

The project has moved from a minimal prototype to a complete applied-track deliverable: runnable system, fixed benchmark, baselines, tests, generated reports, final report, and presentation slides.